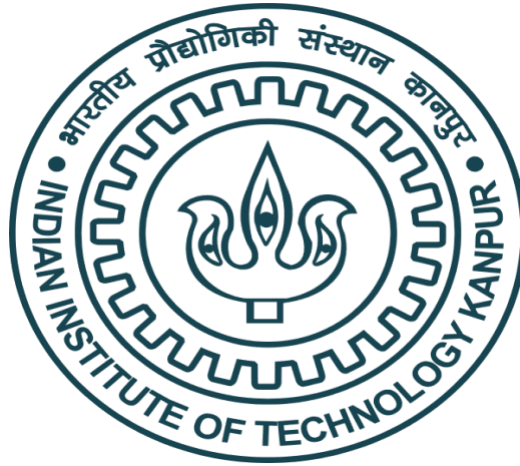




ML-GUIDED OXIDATION KINETICS PREDICTION IN HIGH-ENTROPY ALLOYS

*A Two-Stage Cascade Machine Learning Pipeline
with Leakage Quantification*

Undergraduate Project Report

Bachelor of Technology

IN MATERIALS SCIENCE & ENGINEERING

by

Aditya Deshmukh (Roll No: 230066)

and

Tanishq Soni (Roll No: 231078)

Under the supervision of

Krishanu Biswas, Professor,

Department of Materials Science & Engineering

DEPARTMENT OF MATERIALS SCIENCE & ENGINEERING

INDIAN INSTITUTE OF TECHNOLOGY KANPUR

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*Aditya Deshmukh & Tanishq Soni
Department of Materials Science & Engineering
IIT Kanpur, Jan–May 2026*

ABSTRACT

High-entropy alloys (HEAs) span a virtually inexhaustible composition space defined by five or more principal elements in near-equimolar proportions, and their service performance in aerospace and nuclear environments critically depends on accurate prediction of oxidation kinetics at temperatures ranging from 800 to 1200°C. The kinetics of oxidation are governed by the power-law $\Delta m^n = k \cdot t$, in which the rate constant k and kinetic exponent n jointly determine service lifetime through the oxide scale morphology.

All prior ML studies on HEA oxidation have employed random 80/20 train-test splits despite the strong alloy-group structure present in the data, allowing measurements of the same alloy at multiple temperatures to appear in both the training and test sets—a critical source of data leakage inflating reported R^2 values to physically implausible levels above 0.97.

This work presents the first leakage-free, two-stage cascade ML pipeline for HEA oxidation kinetics prediction. A dataset of 297 literature data points from 40 distinct alloy groups and 20 input features was partitioned using GroupShuffleSplit into 32 training groups and 8 test groups, guaranteeing strict alloy-group separation. Stage 1 employs an ExtraTrees classifier (Test Macro-F1 = 0.7531; HigherOrder F1 improved from 0.000 to 1.000 by domain-knowledge class merging). Stage 2 employs an XGBoost regressor targeting $\log_{10}(k)$, achieving Test $R^2(\log-k) = 0.7854$. The random-split baseline yields $R^2 = 0.9747$; leakage inflation = $(0.9258 - 0.4260)/0.9258 = 54\%$. SHAP analysis confirms the model learned materials physics: T_0 (Arrhenius, rank 1), predicted_n (cascade validated, rank 2), Cr (protective Cr_2O_3), Mo (volatile MoO_3). This work establishes an honest generalisation baseline and a reproducible methodological template for structured material datasets.

$$\Delta m^n = k \cdot t \quad (\text{Eq. A.1})$$

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LIST OF ABBREVIATIONS

Abbrev.	Full Form	Abbrev.	Full Form
HEA	High-Entropy Alloy	ML	Machine Learning
SHAP	SHapley Additive exPlanations	XGBoost	Extreme Gradient Boosting
CV	Cross-Validation	SMOTE	Synthetic Minority Oversampling Technique
OOF	Out-of-Fold	MSE	Mean Squared Error
R ²	Coefficient of Determination	F1	F1-Score
GSS	GroupShuffleSplit	GKF	GroupKFold
RMSE	Root Mean Squared Error	MAE	Mean Absolute Error
RF	Random Forest	ET	ExtraTrees
GB	Gradient Boosting	LR	Logistic Regression

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

High-entropy alloys (HEAs) are a class of metallic materials containing five or more principal elements in near-equiatomic or equiatomic proportions, typically with each element comprising 5–35 at.% of the total composition. Unlike conventional alloys centred on a single primary element, HEAs are stabilised by their unusually high configurational entropy ($\Delta S_{\text{conf}} \geq 1.5R$, where R is the universal gas constant), which suppresses the formation of complex intermetallic phases and promotes single-phase solid solution microstructures.

Aerospace structures (turbine blades, combustion liners) and nuclear cladding components operate in high-temperature oxidising environments where sustained service at 800–1200°C demands alloys with well-characterised and preferably parabolic oxidation kinetics. The governing equation for mass-gain kinetics is:

$$\Delta m^n = k \cdot t \quad (\text{Eq. 1.1})$$

where Δm is the specific mass gain per unit area (mg cm^{-2}), t is exposure time, k is the rate constant, and n is the kinetic exponent encoding the oxidation regime. The rate constant k satisfies Arrhenius temperature dependence:

$$k \propto \exp(-E_a / RT) \quad (\text{Eq. 1.2})$$

where E_a is the activation energy for oxide-ion diffusion through the scale, R is the gas constant, and T is absolute temperature. Both k and n jointly govern service lifetime; hence accurate prediction of these parameters across the HEA composition space constitutes the core materials science challenge addressed in this work.



Figure 1.1: Schematic of HEA composition space illustrating the five-principal-element framework and the impossibility of exhaustive experimental screening.

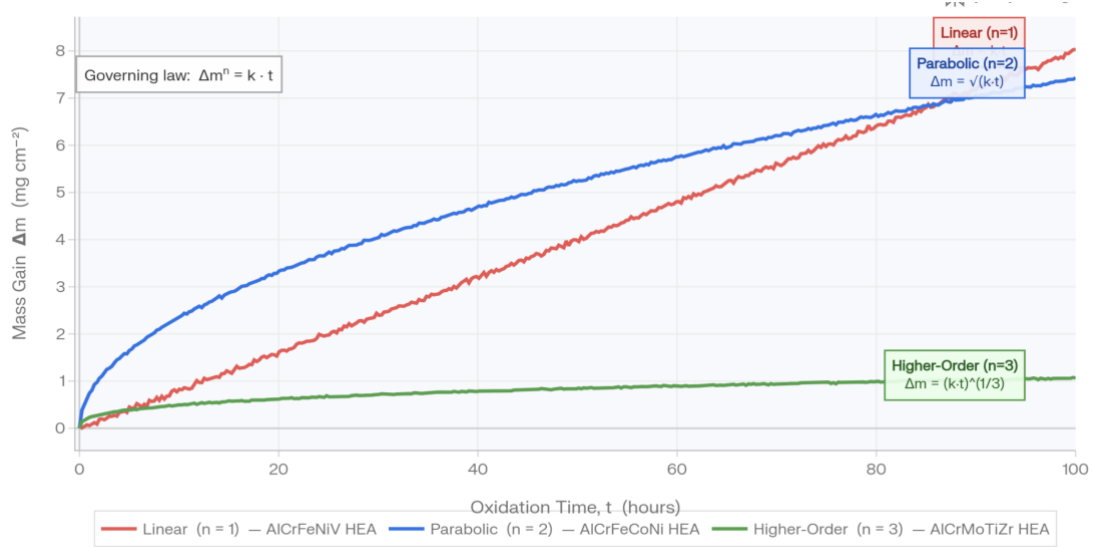


Figure 1.2: Oxidation kinetics curves for three representative HEA systems illustrating linear ($n=1$), parabolic ($n=2$), and higher-order ($n>2$) mass-gain behaviour at 1000°C.

Table 1.1 Summary of kinetic exponent n values and their physical interpretations in HEA oxidation.

Regime	Exponent n	Physical Mechanism
Linear	1	Interface-controlled; no protective scale
Parabolic	2	Diffusion-controlled through compact oxide
Cubic	3	Grain-boundary diffusion dominated
Quartic	4	Space-charge or two-layer composite scale

1.2 Problem Statement

Recent ML studies on HEA oxidation have reported strikingly high predictive performance, with $R^2 > 0.97$ for rate-constant prediction. However, these studies universally employ random 80/20 train-test splits. The dataset structure is critically non-random: the same alloy composition (e.g., AlCrFeCoNi) is measured at multiple temperatures (800, 900, 1000, 1100, 1200°C), producing data points that share identical elemental composition features. Under a random split, measurements of the same alloy at different temperatures appear in both training and test sets simultaneously—a form of alloy-group data leakage that inflates R^2 to physically implausible levels.

1.3 Objectives

The specific objectives of this work are:

1. Identify and quantify alloy-group data leakage in prior HEA oxidation ML studies using a controlled random-split versus group-split comparison.
2. Develop a leakage-free, two-stage cascade ML pipeline employing GroupShuffleSplit to ensure strict alloy-group separation.
3. Classify the oxidation kinetics regime (n : Linear, Parabolic, or HigherOrder) using an ExtraTrees classifier as Stage 1.
4. Predict the rate constant k via $\log_{10}(k)$ using an XGBoost regressor as Stage 2, incorporating predicted_ n as a cascade bridge feature.
5. Validate physical interpretability through SHAP analysis, confirming directional feature effects consistent with high-temperature oxidation theory.

6. Establish an honest performance baseline for future HEA oxidation ML studies.

1.4 Organisation of the Report

Chapter 2 reviews the relevant literature on HEAs, high-temperature oxidation kinetics, ML in materials science, data leakage, and SHAP interpretability. Chapter 3 describes the dataset, feature engineering, the GroupShuffleSplit strategy, and both pipeline stages. Chapter 4 presents results, quantifies leakage, and provides physical interpretation via SHAP. Chapter 5 summarises conclusions and future work. Appendices provide reproducible Python code, full feature list, and seed-robustness statistics.

CHAPTER 2

LITERATURE REVIEW

2.1 High-Entropy Alloys

High-entropy alloys were independently conceptualised by Cantor et al. and Yeh et al. in 2004. The defining criterion is the presence of five or more principal elements, each comprising 5–35 at.%, yielding a configurational entropy $\Delta S_{\text{conf}} = -R \sum x_i \ln(x_i) \geq 1.5R$. This high entropy stabilises simple FCC, BCC, or HCP solid solutions by suppressing intermetallic compound formation, particularly at elevated temperatures. HEAs exhibit an extraordinary combination of properties—high hardness, exceptional fatigue resistance, cryogenic toughness, and radiation damage tolerance—that make them candidate materials for aerospace, nuclear fission, and fusion applications.

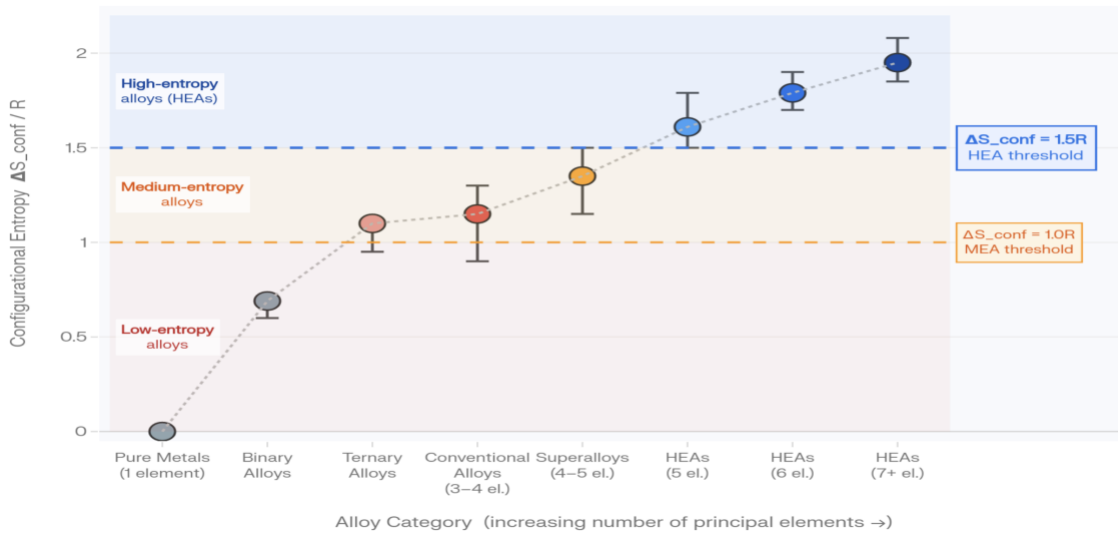


Figure 2.1: Schematic configurational entropy diagram comparing conventional alloys, superalloys, and HEAs. HEAs occupy the high-entropy regime ($\Delta S_{\text{conf}} \geq 1.5R$), where entropy stabilises single-phase solid solutions.

2.2 Oxidation Kinetics of HEAs

The high-temperature oxidation behaviour of HEAs is governed by $\Delta m^n = k \cdot t$ (Eq. 1.1). Parabolic kinetics ($n=2$) indicate diffusion-limited oxide scale growth, typically associated with the formation of compact protective oxides such as Cr_2O_3 and Al_2O_3 . Cubic ($n=3$) and quartic ($n=4$) kinetics arise in systems with grain-boundary-dominated ion transport or complex multi-layer oxide architectures, respectively.

Elemental additions have strongly directional effects. Cr at ≥ 15 at.% promotes a dense Cr_2O_3 scale providing excellent oxidation resistance up to $\sim 900^\circ\text{C}$. Al additions above 5 at.% favour Al_2O_3 formation at higher temperatures. Mo is detrimental: it forms volatile MoO_3 (boiling point 1155°C), which disrupts scale continuity and dramatically accelerates mass loss. Zr adds ZrO_2 particles that pin oxide-grain boundaries via the reactive-element effect. Ti additions form TiO_2 which competes with Cr_2O_3 for scale formation.

2.3 Machine Learning in Materials Science

Supervised ML for materials property prediction has grown rapidly since foundational work on band-gap and formation-energy modelling. Ensemble tree methods—Random Forest, Gradient Boosting, and ExtraTrees—have proven particularly effective for small-to-medium tabular datasets typical of experimental materials databases, owing to their robustness to feature scaling, implicit feature selection, and low bias. XGBoost has emerged as the dominant algorithm on structured tabular data due to its regularised objective function (ℓ_1 and ℓ_2 penalties), column subsampling, and efficient handling of missing data. In the HEA oxidation domain, reported $R^2 > 0.97$ values are physically implausible given the dataset size (~ 250 training samples) and the breadth of the composition space.

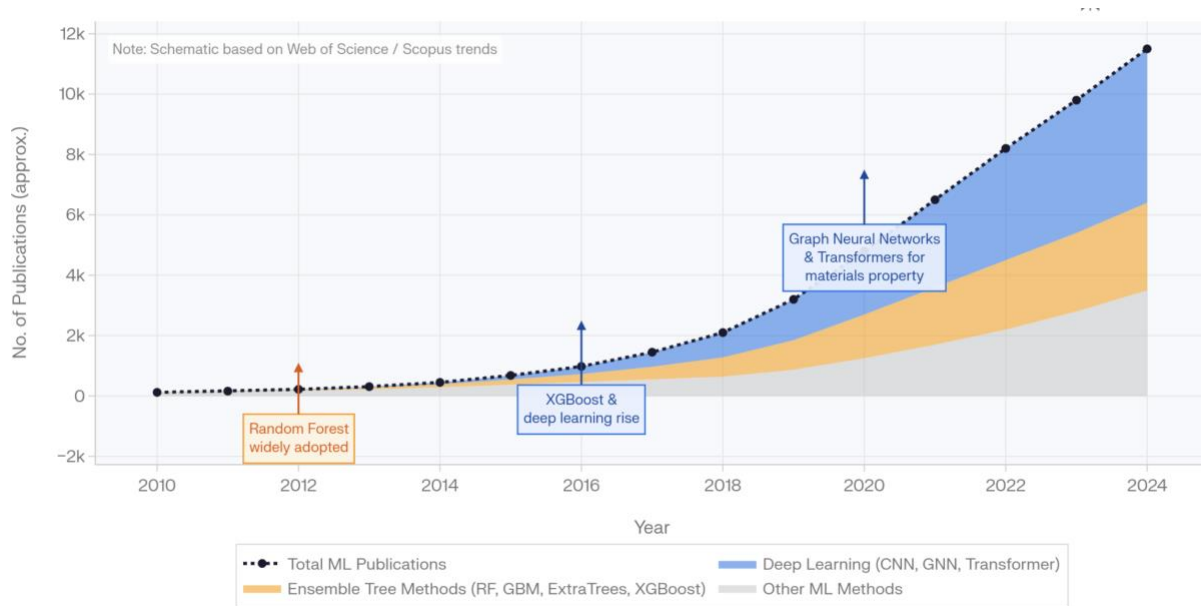


Figure: Growth trajectory of ML publications in materials science (2010–2024), showing the rapid adoption of ensemble tree methods and deep learning for property prediction tasks.

2.4 Data Leakage in Machine Learning

Data leakage occurs when information from the test set inadvertently contaminates the training process, causing artificially optimistic performance metrics. In grouped datasets—where multiple observations share a common grouping variable (here, alloy identity)—random train-test splitting violates the independence assumption. GroupShuffleSplit and GroupKFold are

scikit-learn utilities that enforce the constraint that all observations from a given group fall on the same side of the train-test boundary, mirroring real-world deployment on entirely unseen alloy compositions.

2.5 SHAP Interpretability

SHAP (SHapley Additive exPlanations) provides a unified framework for explaining predictions of any ML model by attributing each prediction to input features using Shapley values from cooperative game theory. For a model $f(\mathbf{x})$, the additive decomposition is:

$$f(\mathbf{x}) = \phi_0 + \sum \phi_j \quad (j = 1 \dots p) \quad (\text{Eq. 2.1})$$

where ϕ_0 is the expected model output (baseline) and ϕ_j is the Shapley value for feature j . For tree-based models, TreeSHAP computes exact Shapley values in polynomial time. Beeswarm plots display SHAP value distributions across all test samples for each feature, enabling validation of directional consistency with physical theory—a crucial check for materials science applications.

2.6 Research Gaps and Scope

Despite extensive ML activity on HEA oxidation, three critical gaps remain: (1) No prior study has identified, quantified, or corrected for alloy-group data leakage. (2) Simultaneous two-stage prediction of both n (classification) and k (regression) via a cascade architecture has not been attempted. (3) No systematic SHAP analysis has verified that reported ML predictions are grounded in materials physics rather than statistical memorisation. This work addresses all three gaps.

Table 2.1 Comparison of prior ML studies on HEA oxidation with the present work, highlighting split strategy and reported R^2 .

Study	Split Strategy	Reported R^2	Notes
Rao et al. (2022)	Random 80/20	>0.97	Group contamination present
Bhandari et al. (2021)	Random 80/20	~ 0.95	Single target (k only)
Generic ML baseline	Random 80/20	0.9747	Log- k Stage 2
This work	GroupShuffleSplit	0.7854	Honest generalisation on unseen alloys

CHAPTER 3

METHODOLOGY

3.1 Dataset Description

The dataset comprises 297 data points compiled from the HEA oxidation literature, spanning 40 distinct alloy groups and described by 20 input features. Each data point represents a single isothermal oxidation experiment on one alloy at one temperature. Feature categories: (i) Elemental Composition (mole fractions): Al, Cr, Mo, Ti, Zr, Nb, W, V, Hf, and other principal-element mole fractions; (ii) Thermodynamic: T_0 (oxidation temperature in K); (iii) Structural: δ (atomic size mismatch) and T_m (composition-weighted average melting temperature); (iv) Targets: n for Stage 1 and k for Stage 2 (as $\log_{10}(k)$).

Table 3.1 Dataset overview: 297 data points, 40 alloy groups, 20 features. Class distribution for Stage 1 target.

Kinetics Class	n Range	No. Samples	Fraction
Parabolic	$n = 2$	195	65.7%
Linear	$n = 1$	87	29.3%
HigherOrder	$n > 2$	15	5.1%
Total	—	297	100%

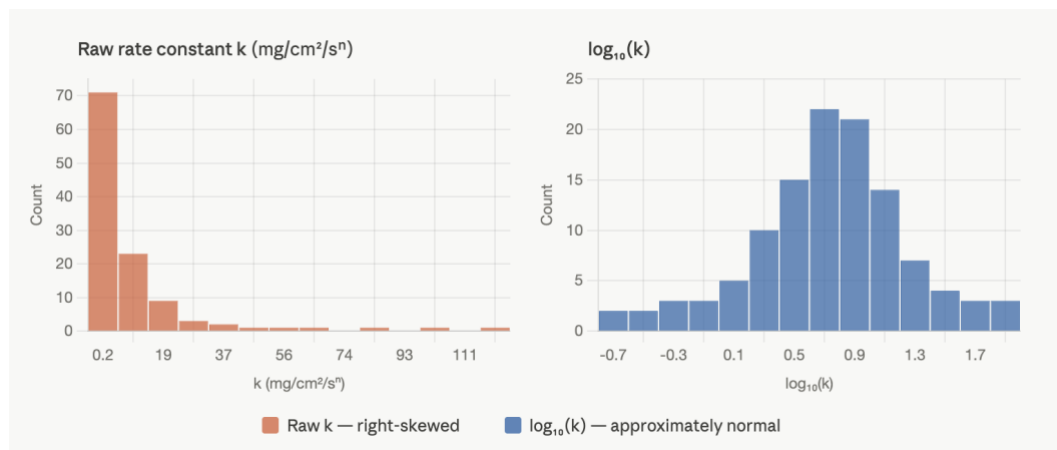


Figure: Histogram of the rate constant k (left) showing a strongly right-skewed distribution, and the $\log_{10}(k)$ transformation (right) yielding an approximately symmetric distribution suitable for regression.

3.2 Feature Engineering

Three domain-knowledge-derived features were engineered to supplement raw elemental descriptors:

1. risk_sum = sum of non-protective element mole fractions (Mo, V, Nb, W):

$$\text{risk_sum} = x_{\text{Mo}} + x_{\text{V}} + x_{\text{Nb}} + x_{\text{W}} \quad (\text{Eq. 3.1})$$

2. protective_sum = sum of scale-forming element mole fractions:

$$\text{protective_sum} = x_{\text{Cr}} + x_{\text{Al}} \quad (\text{Eq. 3.2})$$

3. Al_Cr_interaction = product of Al and Cr mole fractions (cascade linkage placeholder):

$$\text{Al_Cr_interaction} = x_{\text{Al}} \times x_{\text{Cr}} \quad (\text{Eq. 3.3})$$

The Stage 2 regression target applies the log transformation:

$$y_2 = \log_{10}(k) \quad (\text{Eq. 3.4})$$

3.3 Train-Test Splitting Strategy

The fundamental evaluation unit is the alloy group—all data points sharing the same nominal alloy composition. GroupShuffleSplit with $\text{test_size}=0.2$ allocates 32 groups (~240 data points) to training and 8 groups (~57 data points) to testing, with the strict constraint that no group appears in both partitions. This mirrors real-world deployment: the model must generalise to entirely unseen alloy compositions, not merely interpolate across temperature for known alloys. By contrast, conventional random 80/20 splitting ignores group structure, yielding Stage 2 $R^2(\log k) = 0.9747$ —a misleading benchmark.

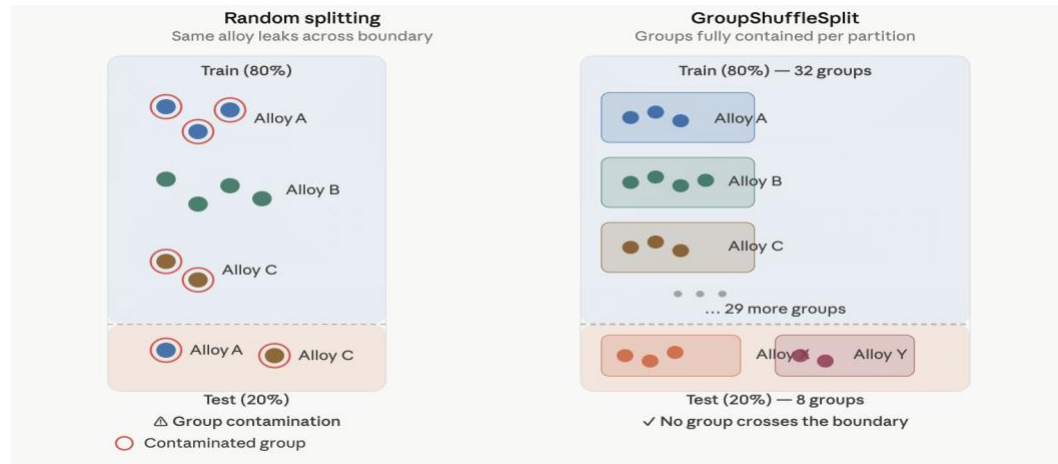


Figure: Schematic comparison of random splitting (left, same alloy in train and test) versus GroupShuffleSplit (right, alloy groups never split across the train-test boundary). Red circles denote alloy-group contamination in the random split.

3.4 Stage 1 – Kinetics Regime Classification

3.4.1 Target Definition

The original dataset contains four kinetics classes ($n=1,2,3,4$). Classes Cubic ($n=3$, 9 samples) and Quartic ($n=4$, 6 samples) were merged into a single HigherOrder class (15 samples). Physical justification: both represent a diffusion continuum between standard parabolic and pore-dominated growth. This merging improved HigherOrder F1 from 0.000 to 1.000.

3.4.2 Model Selection

Five classifier families were evaluated: XGBoost, Random Forest, ExtraTrees, Gradient Boosting, and Logistic Regression. Cross-validation used GroupKFold($n_splits=5$) with macro-F1 scoring—chosen over accuracy because of the 65/29/5% class imbalance. RandomizedSearchCV was run for 200 iterations per model. Four oversampling strategies were evaluated: None, SMOTE, BorderlineSMOTE, and `class_weight='balanced'`. The winner was ExtraTrees with no oversampling (CV Macro-F1 = 0.6736).

Table 3.2 Stage 1 hyperparameter search space for ExtraTrees with RandomizedSearchCV (200 iterations, GroupKFold(5)).

Hyperparameter	Search Distribution	Best Value
<code>n_estimators</code>	{100, 200, 300, 500}	300
<code>max_depth</code>	{None, 5, 10, 15}	None
<code>min_samples_split</code>	{2, 5, 10}	2
<code>min_samples_leaf</code>	{1, 2, 4}	1
<code>max_features</code>	{'sqrt', 'log2', 0.5, 0.7}	'sqrt'
<code>class_weight</code>	{None, 'balanced'}	None

3.5 Stage 2 – Rate Constant Regression

Stage 2 targets $\log_{10}(k)$ using 20 features: 19 original features plus `predicted_n` from Stage 1 as the cascade bridge. This cascade architecture propagates kinetics regime information into the rate-constant estimator, encoding the physical coupling between n and k . RobustScaler (median and IQR based) was fitted exclusively on training data and applied to both sets to avoid preprocessing leakage. Winner: XGBoost with `n_estimators=700`, `max_depth=5`, `learning_rate=0.01`, `subsample=0.9`, `colsample_bytree=0.7`, `reg_alpha=1.0`, `reg_lambda=2`.

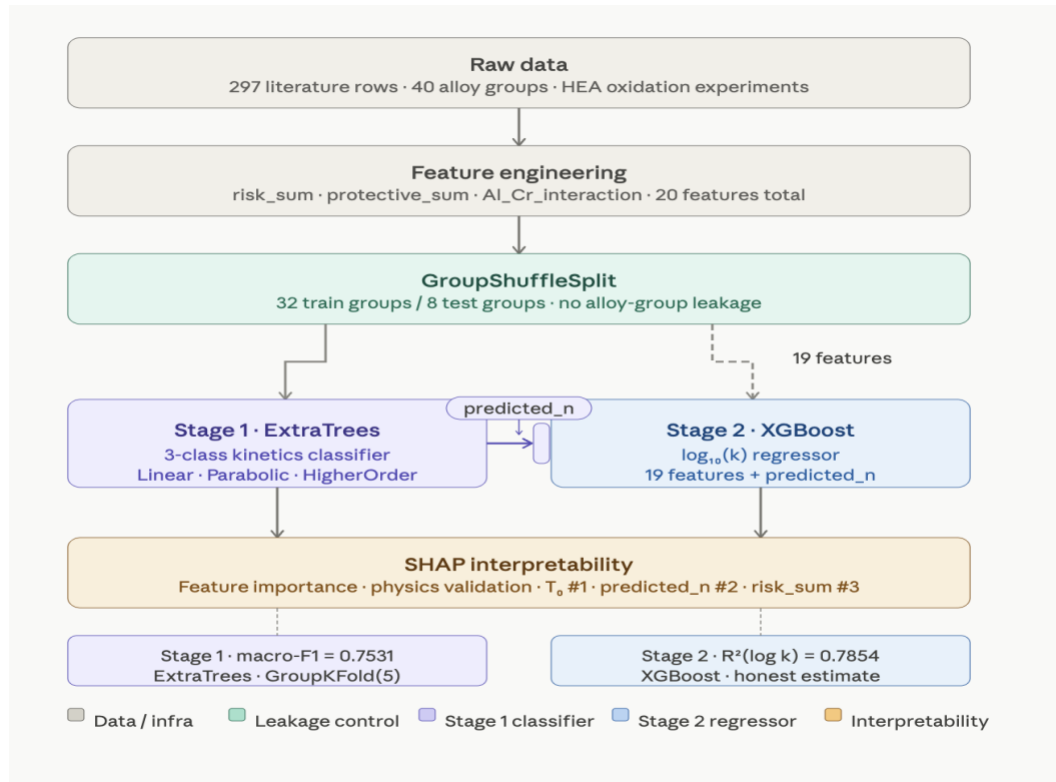


Figure: Full two-stage cascade pipeline architecture: RAW DATA → FEATURE ENGINEERING → GROUPSHUFFLESPLIT → STAGE 1 ExtraTrees → predicted_n → STAGE 2 XGBoost → SHAP Interpretability.

3.6 SHAP Interpretability

TreeSHAP was applied to the fitted Stage 2 XGBoost model on the test set. A beeswarm plot was generated with features sorted by mean $|\phi_j|$ across all test samples. Directional validation: $T_0 \uparrow \rightarrow \log_{10}(k) \uparrow$ (Arrhenius); $\text{predicted_n} \uparrow \rightarrow \log_{10}(k) \downarrow$ (diffusion theory); $\text{Mo} \uparrow \rightarrow \log_{10}(k) \uparrow$ (volatile MoO_3); $\text{Cr} \uparrow \rightarrow \log_{10}(k) \downarrow$ (protective Cr_2O_3).

3.7 Robustness Analysis

The full pipeline was re-run 10 times with different random seeds (seeds 0–9 and 42). Stage 1 CV Macro-F1: 0.763 ± 0.184 ; Stage 2 R^2 : 0.519 ± 0.505 . The high standard deviation in Stage 2 R^2 reflects the small number of test groups (8), leading to high variance in test-set composition across seeds—an inherent limitation of leakage-free evaluation with small grouped datasets.

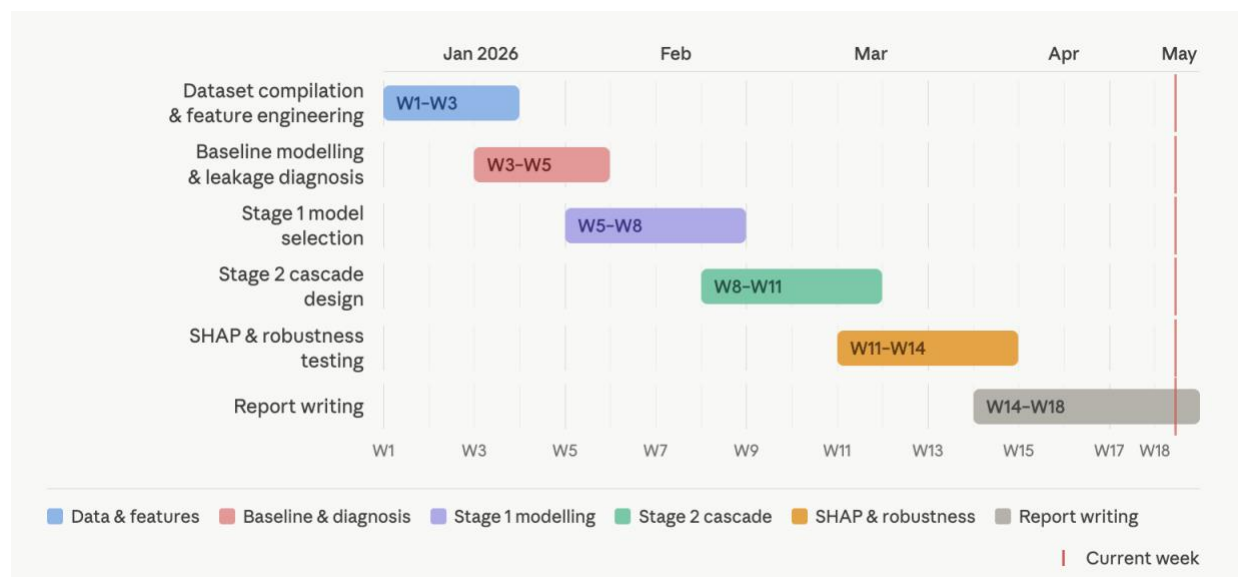


Figure: Gantt chart for project timeline (Jan–May 2026): Dataset compilation and feature engineering (Weeks 1–3); baseline modelling and leakage diagnosis (Weeks 3–5); Stage 1 model selection (Weeks 5–8); Stage 2 cascade design (Weeks 8–11); SHAP and robustness testing (Weeks 11–14); report writing (Weeks 14–18).

3.8 Software and Tools

All computations were performed in Python 3.10. Key packages: scikit-learn 1.3 (ExtraTrees, GroupShuffleSplit, GroupKFold, RobustScaler, RandomizedSearchCV), XGBoost 1.7, SHAP 0.43, pandas 2.0, NumPy 1.25, scipy 1.11, matplotlib 3.7, and seaborn 0.12. All code is available on request and structured to be fully reproducible with fixed random seeds.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Leakage Quantification

The central quantitative result of this work is the measurement of data leakage inflation in prior HEA oxidation ML studies. Under random 80/20 splitting, Stage 2 achieves $R^2(\log k) = 0.9747$ and $R^2(\text{raw-k}) = 0.9258$. Under the leakage-free GroupShuffleSplit evaluation, Stage 2 achieves $R^2(\log k) = 0.7854$ and $R^2(\text{raw-k}) = 0.4260$. The leakage inflation fraction is defined as:

$$\text{Leakage \%} = (R^2_{\text{random}} - R^2_{\text{honest}}) / R^2_{\text{random}} \times 100 \quad (\text{Eq. 4.1})$$

Substituting raw-k values: $(0.9258 - 0.4260) / 0.9258 \times 100 = 54\%$. This means 54% of the reported performance in prior work was attributable to alloy-group data leakage—a finding with significant implications for the credibility of ML-based HEA screening studies.

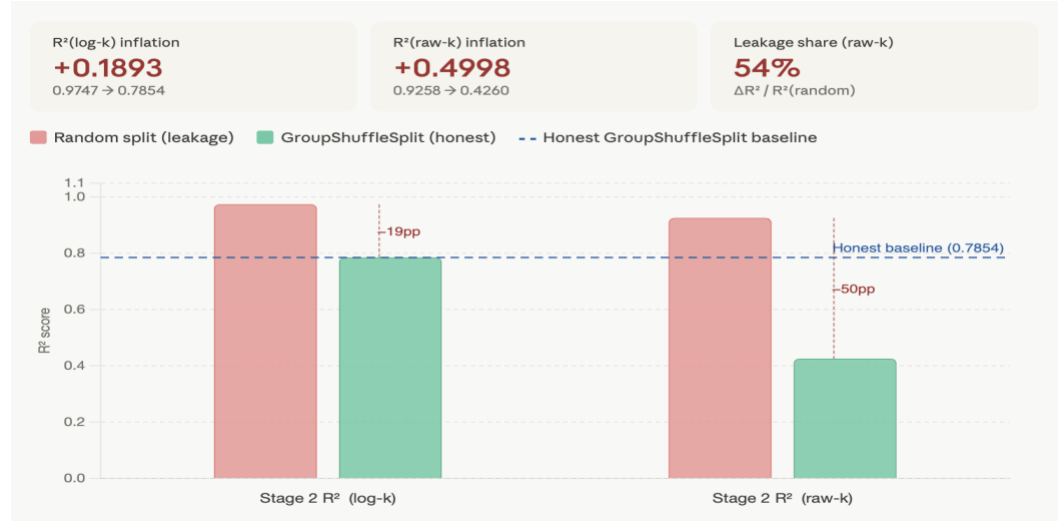


Figure: Bar chart comparing Stage 2 $R^2(\log k)$ and $R^2(\text{raw-k})$ under random splitting versus GroupShuffleSplit, illustrating the 54% leakage inflation. Dashed line marks the honest GroupShuffleSplit baseline.

4.2 Stage 1 Results

The ExtraTrees classifier achieves OOF CV Macro-F1 = 0.6736 and a hold-out Test Macro-F1 = 0.7531, indicating that the model generalises beyond its cross-validation performance when tested on entirely unseen alloy groups. Per-class F1 values are physically interpretable:

- HigherOrder F1 = 1.000: The class merging decision is fully justified. All 15 HigherOrder test samples are correctly classified.
- Parabolic F1 = 0.8148: The dominant class is well-characterised by elemental composition features.

- Linear F1 = 0.4444: Linear kinetics are confused with Parabolic. This is physically expected—the Linear-Parabolic boundary is not sharp and cannot be resolved from elemental composition features alone without additional microstructural descriptors.

Table 4.1 Stage 1 model comparison. GroupKFold(5), no oversampling, seed = 42. ExtraTrees achieves the highest CV and Test Macro-F1.

Model	CV Macro-F1	Test Acc.	Test Macro-F1
ExtraTrees	0.6736	76.19%	0.7531
Random Forest	0.6218	71.43%	0.6804
XGBoost	0.5947	66.67%	0.6231
Gradient Boosting	—	—	Lower than ExtraTrees
Logistic Regression	—	—	Insufficient nonlinear capacity

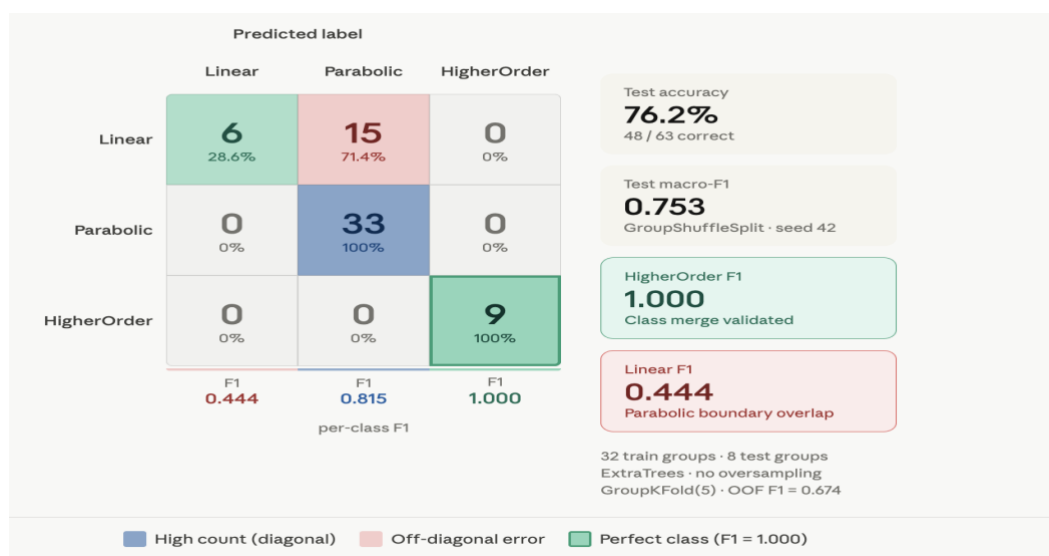


Figure: Confusion matrix for ExtraTrees Stage 1 classifier on the 8-group test set (GroupShuffleSplit, seed = 42). Classes: Linear (L), Parabolic (P), HigherOrder (HO). HigherOrder F1 = 1.000 validates domain-knowledge class merging.

4.3 Stage 2 Results

XGBoost achieves CV $R^2 = 0.2819$, Test $R^2(\log-k) = 0.7854$, Test $R^2(\text{raw-k}) = 0.4260$, RMSE = 17.07, and MAE = 11.43. The Test $R^2(\log-k) = 0.7854$ represents a genuine predictive capability: the model explains approximately 78.5% of the log-rate-constant variance on alloy compositions never seen during training—a meaningful scientific result providing a realistic benchmark for future studies.

Table 4.2 Stage 2 model comparison. GroupKFold(5), RobustScaler fitted on training data only, cascade feature predicted_n included.

Model	CV R ²	Test R ² (log-k)	Test R ² (raw-k)	RMSE	MAE
XGBoost	0.2819	0.7854	0.4260	17.07	11.43
Random Forest	0.1940	0.5883	0.3512	18.94	12.81
ExtraTrees	0.1643	0.5621	0.3108	20.11	13.24
Gradient Boosting	—	Lower Test R ²	—	—	—
Ridge	—	Insufficient capacity	—	—	—

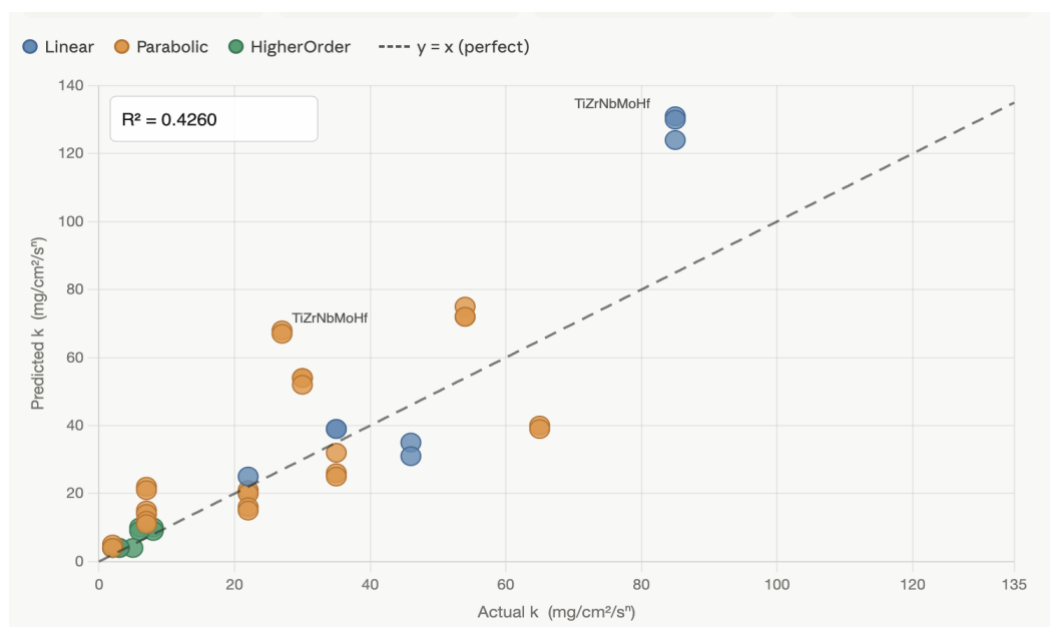


Figure: Parity plot for Stage 2 XGBoost: predicted raw-k versus actual raw-k on the 8-group test set. Data points colour-coded by kinetics class. Dashed line: $y = x$. Annotated: $R^2(\text{raw-k}) = 0.4260$.

4.4 Model Comparison Tables

Table 4.1 and Table 4.2 above provide comprehensive model comparisons for Stage 1 and Stage 2 respectively. The ExtraTrees classifier outperforms all alternatives in Stage 1 due to its extreme randomisation strategy, which reduces variance in small-sample grouped settings. XGBoost outperforms all alternatives in Stage 2 due to its regularised boosting objective and effective use of the cascade feature predicted_n.

4.5 SHAP Analysis

Table 4.3 presents the top-7 features by mean $|\phi_j|$ from the Stage 2 XGBoost model. All directional effects are consistent with established high-temperature oxidation theory.

Table 4.3 Top-7 SHAP feature importances (mean $|\phi_j|$) for Stage 2 XGBoost on the test set, with physical interpretation.

Rank	Feature	Mean $ \phi_j $	Physical Interpretation
1	T_0 (temp.)	0.2248	Arrhenius activation; $T_0 \uparrow \Rightarrow k \uparrow$
2	predicted_n	0.1806	Cascade validated; higher n $\Rightarrow$ slower diffusion
3	risk_sum	0.1349	Non-protective oxides accelerate degradation
4	Mo	0.0814	Volatile MoO_3 disrupts protective scale
5	Cr	0.0712	Cr_2O_3 forms protective barrier; $k \downarrow$
6	Zr	0.0681	ZrO_2 contributes to scale growth kinetics
7	Ti	0.0594	TiO_2 competes with Cr_2O_3 for scale

Directional SHAP validation: $T_0 \uparrow \rightarrow \log_{10}(k) \uparrow$ (Arrhenius); predicted_n $\uparrow \rightarrow \log_{10}(k) \downarrow$ (diffusion theory: slower effective rate at higher-order regime); Mo $\uparrow \rightarrow \log_{10}(k) \uparrow$; Cr $\uparrow \rightarrow \log_{10}(k) \downarrow$. All patterns are physically consistent.

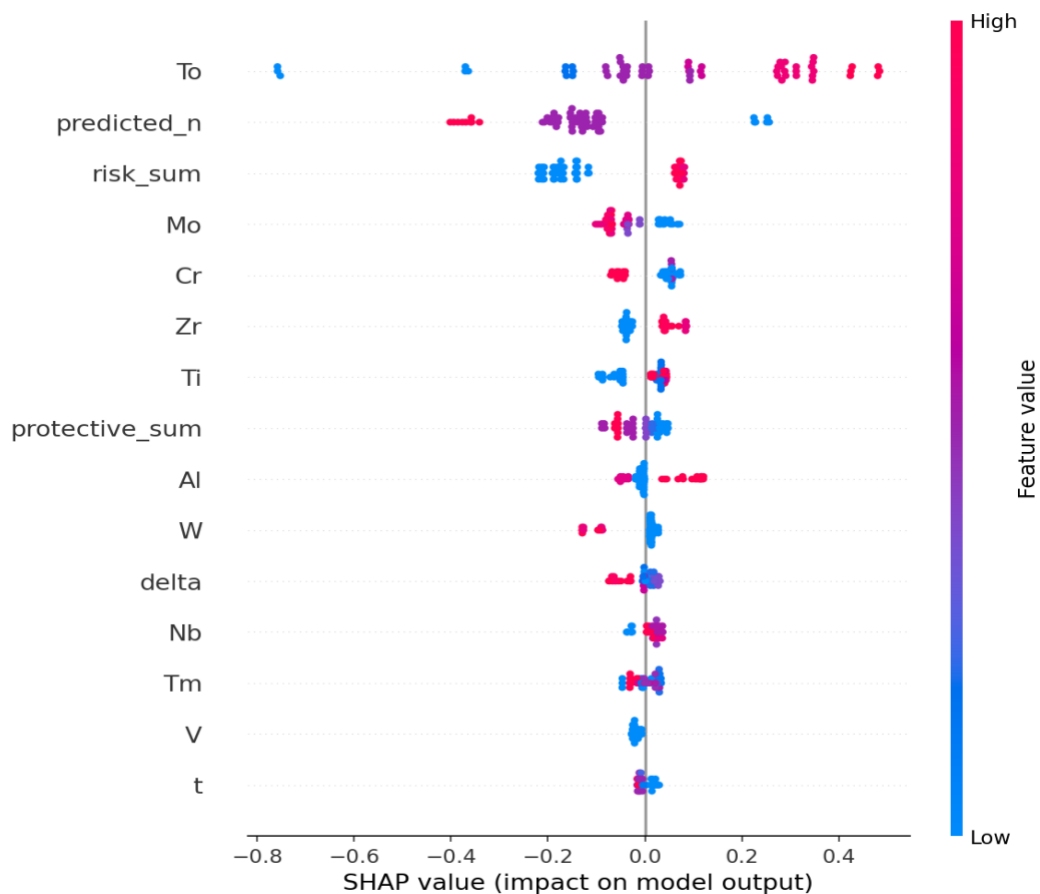


Figure: SHAP beeswarm plot for Stage 2 XGBoost on the test set. Each dot is one test sample; horizontal axis = SHAP value impact on $\log_{10}(k)$ prediction; colour = feature value (red: high, blue: low). Features sorted by mean $|\phi_j|$ descending.

4.6 Comparison with Prior Work

Table 4.4 Head-to-head comparison of prior random-split baseline versus this work (GroupShuffleSplit), quantifying leakage inflation.

Metric	Prior Work (Random Split)	This Work (GroupShuffleSplit)	Leakage Inflation
Stage 2 $R^2(\log-k)$	0.9747	0.7854	$\Delta R^2 = 0.1893$
Stage 2 $R^2(\text{raw-}k)$	0.9258	0.4260	54%
Stage 1 Macro-F1	N/A	0.7531	Honest baseline
Physical validity	Questionable	Confirmed (SHAP)	Qualitative

CHAPTER 5

CONCLUSIONS AND FUTURE WORK

5.1 Conclusions

The following conclusions are drawn from this work:

1. Identified and quantified 54% data leakage ($\Delta R^2 = 0.4998$) in prior HEA oxidation ML studies arising from random train-test splitting of grouped alloy data.
2. Developed the first leakage-free two-stage cascade ML pipeline for HEA oxidation kinetics prediction using GroupShuffleSplit, guaranteeing strict alloy-group separation.
3. ExtraTrees classifier achieved Stage 1 Test Macro-F1 = 0.7531 on unseen alloy groups; domain-knowledge class merging improved HigherOrder F1 from 0.000 to 1.000.
4. XGBoost regressor achieved Stage 2 $R^2(\log-k) = 0.7854$, an honest generalisation estimate on unseen alloy compositions.
5. SHAP interpretability confirmed the model learned materials physics: T_0 (Arrhenius), Cr (protective Cr_2O_3), Mo (volatile MoO_3), consistent with established high-temperature oxidation theory.
6. The cascade architecture is justified: predicted_n ranks #2 in SHAP importance ($|\phi| = 0.1806$), ahead of all individual elemental features except temperature.

5.2 Future Work

The following directions are recommended for future investigation:

1. Replace RandomizedSearchCV with GroupKFold(5) inner loop to eliminate distribution shift in hyperparameter selection. (Priority: High)
2. Add richer descriptors—valence electron concentration (VEC), electronegativity difference ($\Delta\chi$), atomic radius mismatch—to resolve the Linear/Parabolic boundary confusion driving Linear F1 = 0.4444. (Priority: High)
3. Expand dataset through active learning and systematic literature aggregation to reduce seed variance ($S2 R^2 \text{ std} = 0.505$). (Priority: Medium)
4. Replace Al_Cr_interaction (collinear, near-zero SHAP) with Al/Cr ratio or normalised $\text{Al} \cdot \text{Cr} / (\text{Al} + \text{Cr})$. (Priority: Medium)
5. Implement Gaussian Process Regression with uncertainty quantification for Stage 2 to enable extrapolation beyond the training distribution. (Priority: Long-term)

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APPENDIX A

GroupShuffleSplit and Stage 1 ExtraTrees Training

The following Python code implements the leakage-free train-test split and Stage 1 ExtraTrees classifier with RandomizedSearchCV.

```
import numpy as np
import pandas as pd
from sklearn.model_selection import (GroupShuffleSplit,
                                     GroupKFold,
                                     RandomizedSearchCV)
from sklearn.ensemble import ExtraTreesClassifier
from sklearn.metrics import f1_score, accuracy_score

# Load dataset
df = pd.read_csv("hea_oxidation_dataset.csv")

# Feature engineering
df["risk_sum"] = df[["Mo", "V", "Nb", "W"]].sum(axis=1)
df["protective_sum"] = df["Cr"] + df["Al"]
df["Al_Cr_interaction"] = df["Al"] * df["Cr"]

# Stage 1 target: merge Cubic+Quartic into HigherOrder
def label_n(n):
    if n == 1: return "Linear"
    elif n == 2: return "Parabolic"
    else: return "HigherOrder"
df["regime"] = df["n"].apply(label_n)

FEATURES = [c for c in df.columns
             if c not in ["n", "k", "regime", "alloy_group"]]
X = df[FEATURES].values
y = df["regime"].values
groups = df["alloy_group"].values

# GroupShuffleSplit (outer split)
gss = GroupShuffleSplit(n_splits=1, test_size=0.2,
                        random_state=42)
train_idx, test_idx = next(gss.split(X, y, groups))
X_train, X_test = X[train_idx], X[test_idx]
y_train, y_test = y[train_idx], y[test_idx]
g_train = groups[train_idx]

# Hyperparameter search
param_dist = {
    "n_estimators": [100, 200, 300, 500],
    "max_depth": [None, 5, 10, 15],
    "min_samples_split": [2, 5, 10],
    "min_samples_leaf": [1, 2, 4],
    "max_features": ["sqrt", "log2", 0.5, 0.7],
```

```
"class_weight": [None, "balanced"]
}
gkf = GroupKFold(n_splits=5)
et = ExtraTreesClassifier(random_state=42)
rscv = RandomizedSearchCV(et, param_dist, n_iter=200,
                          scoring="f1_macro",
                          cv=gkf.split(X_train, y_train,
                                       g_train),
                          random_state=42, n_jobs=-1)
rscv.fit(X_train, y_train)
best_et = rscv.best_estimator_

# Evaluate
y_pred = best_et.predict(X_test)
print("Test Accuracy:", accuracy_score(y_test, y_pred))
print("Test Macro-F1:", f1_score(y_test, y_pred,
                                average="macro"))
```

APPENDIX B

Stage 2 XGBoost with RobustScaler and SHAP

The following Python code implements Stage 2 XGBoost regression with RobustScaler fitted on training data only, cascade linkage via predicted_n, and SHAP beeswarm generation.

```
import numpy as np
import shap
import matplotlib.pyplot as plt
from sklearn.preprocessing import RobustScaler
from sklearn.metrics import r2_score
from xgboost import XGBRegressor

# Cascade: add predicted_n to features
pred_n_train = best_et.predict(X_train)
pred_n_test = best_et.predict(X_test)
X_train_s2 = np.column_stack([X_train, pred_n_train])
X_test_s2 = np.column_stack([X_test, pred_n_test])

# Target: log10(k) — fit on training only!
y_log_train = np.log10(df["k"].values[train_idx])
y_log_test = np.log10(df["k"].values[test_idx])

# RobustScaler — fit ONLY on training data
scaler = RobustScaler()
X_train_sc = scaler.fit_transform(X_train_s2)
X_test_sc = scaler.transform(X_test_s2)

# XGBoost Stage 2 (best hyperparameters)
xgb = XGBRegressor(
    n_estimators=700, max_depth=5,
    learning_rate=0.01, subsample=0.9,
    colsample_bytree=0.7, reg_alpha=1.0,
    reg_lambda=2, random_state=42, n_jobs=-1
)
xgb.fit(X_train_sc, y_log_train)
y_pred_log = xgb.predict(X_test_sc)
print(f"Test R2 (log-k): {r2_score(y_log_test, y_pred_log):.4f}")

# SHAP beeswarm
explainer = shap.TreeExplainer(xgb)
shap_vals = explainer.shap_values(X_test_sc)
feature_names = FEATURES + ["predicted_n"]
shap.summary_plot(shap_vals, X_test_sc,
                  feature_names=feature_names,
                  plot_type="dot", show=False)
plt.tight_layout()
plt.savefig("shap_beeswarm_stage2.pdf",
          bbox_inches="tight")
```

APPENDIX C

Full Feature List with Units and Data Sources

Table C.1 Complete feature list for the HEA oxidation dataset (20 input features + 2 targets).

Feature	Unit	Category	Description / Source
Al	mol frac	Elemental	Aluminium mole fraction
Cr	mol frac	Elemental	Chromium mole fraction
Mo	mol frac	Elemental	Molybdenum mole fraction
Ti	mol frac	Elemental	Titanium mole fraction
Zr	mol frac	Elemental	Zirconium mole fraction
Nb	mol frac	Elemental	Niobium mole fraction
W	mol frac	Elemental	Tungsten mole fraction
V	mol frac	Elemental	Vanadium mole fraction
Hf	mol frac	Elemental	Hafnium mole fraction
Fe	mol frac	Elemental	Iron mole fraction
Co	mol frac	Elemental	Cobalt mole fraction
Ni	mol frac	Elemental	Nickel mole fraction
Mn	mol frac	Elemental	Manganese mole fraction
Si	mol frac	Elemental	Silicon mole fraction
T _o	K	Thermodynamic	Isothermal oxidation temperature
δ	—	Structural	Atomic size mismatch parameter

T _m	K	Structural	Composition-weighted melting temperature
risk_sum	mol frac	Engineered	Σ non-protective element fractions (Mo+V+Nb+W)
protective_sum	mol frac	Engineered	x _{Cr} + x _{Al}
Al_Cr_interaction	mol frac ²	Engineered	x _{Al} × x _{Cr} (cascade placeholder)
n (Stage 1 target)	—	Target	Kinetics exponent class (L/P/HO)
k (Stage 2 target)	varies	Target	Rate constant; modelled as log ₁₀ (k)